

On the generalized Bohnenblust–Hille inequality for real scalars

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Abstract In this paper we obtain new lower and upper estimates for the sharp constants in the generalized Bohnenblust–Hille inequality introduced in Albuquerque et al. (J Funct Anal 266:3726–3740, 2014). We apply these results to find optimal constants in the generalized Bohnenblust–Hille inequality and also to recover the optimal constants of the mixed (ℓ_1, ℓ_2) -Littlewood inequalities recently obtained in Pellegrino (J Number Theory 160:11–18, 2016) and Pellegrino and Teixeira (Commun Contemp Math, to appear).

Keywords Bohnenblust–Hille inequality · Mixed (ℓ_1, ℓ_2) -Littlewood inequality · Multilinear norms

Mathematics Subject Classification 11Y60

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1 Introduction

The multilinear Bohnenblust–Hille inequality [4] for $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$ asserts that there exists a sequence of positive scalars $(B_m^{\mathbb{K}})_{m=1}^{\infty}$ in $[1, \infty)$ such that

$$\left(\sum_{i_1, \dots, i_m=1}^{\infty} |U(e_{i_1}, \dots, e_{i_m})|^{\frac{2m}{m+1}} \right)^{\frac{m+1}{2m}} \leq B_m^{\mathbb{K}} \|U\| \quad (1)$$

for all continuous m -linear forms $U : c_0 \times \dots \times c_0 \rightarrow \mathbb{K}$, where $(e_i)_{i=1}^{\infty}$ denotes the sequence of canonical vectors of c_0 .

A crucial tool to prove the Bohnenblust–Hille inequality for multilinear forms is the so called *mixed (ℓ_1, ℓ_2) -Littlewood inequality* which asserts (in the real case) that for all continuous m -linear forms $U : c_0 \times \dots \times c_0 \rightarrow \mathbb{R}$ we have

$$\sum_{j_1=1}^{\infty} \left(\sum_{j_2, \dots, j_m=1}^{\infty} |U(e_{j_1}, \dots, e_{j_m})|^2 \right)^{\frac{1}{2}} \leq (\sqrt{2})^{m-1} \|U\|, \quad (2)$$

where, from now on, $\|U\|$ denotes the sup-norm of U . From this inequality and using the intrinsic symmetry of the context, it is not difficult to prove, using a Minkowski-type inequality (see [7, Corollary 5.4.2]), that in fact for each $k \in \{2, \dots, m\}$ we have

$$\begin{aligned} & \left(\sum_{j_1, \dots, j_{k-1}=1}^{\infty} \left(\sum_{j_k=1}^{\infty} \left(\sum_{j_{k+1}, \dots, j_m=1}^{\infty} |U(e_{j_1}, \dots, e_{j_m})|^2 \right)^{\frac{1}{2} \times 1} \right)^{\frac{1}{1} \times 2} \right)^{\frac{1}{2}} \\ & \leq (\sqrt{2})^{m-1} \|U\|, \end{aligned} \quad (3)$$

for all continuous m -linear forms $U : c_0 \times \dots \times c_0 \rightarrow \mathbb{R}$, which is also called mixed (ℓ_1, ℓ_2) -Littlewood inequality. In a more simplified notation we have m inequalities with “multiple” exponents $(1, 2, 2, \dots, 2), \dots, (2, \dots, 2, 1)$.

When we replace $\mathbb{R}$ by $\mathbb{C}$ it is well known that $(\sqrt{2})^{m-1}$ can be replaced by $(2/\sqrt{\pi})^{m-1}$.

The connections between the mixed (ℓ_1, ℓ_2) -Littlewood inequality and the Bohnenblust–Hille inequality are well known and can be easily explained through the interpolative approach from [1, Sect. 2]. In fact, in order to obtain the Bohnenblust–Hille inequality from the mixed (ℓ_1, ℓ_2) -Littlewood inequalities it suffices to observe that the exponent $\frac{2m}{m+1}$ can be seen as a multiple exponent $\left(\frac{2m}{m+1}, \dots, \frac{2m}{m+1}\right)$ and this exponent is precisely the interpolation of the exponents

$$(1, 2, 2, \dots, 2), \dots, (2, \dots, 2, 1) \quad (4)$$

with $\theta_1 = \dots = \theta_m = 1/m$.

It was proved in [12] that for real scalars the values $(\sqrt{2})^{m-1}$ are the sharp constants for (2). Unfortunately, the proof of [12] cannot be imitated to cover the general family of inequalities (3). Of course, a natural question, whose answer can be useful in other inequalities, is whether the upper estimates $(\sqrt{2})^{m-1}$ can be improved as long as the exponent 1 moves from the left to the right in (4). That question was recently answered by Pellegrino and Teixeira in [13]; in fact the authors show that the sharp constants for any mixed (ℓ_1, ℓ_2) -Littlewood inequality are $(\sqrt{2})^{m-1}$.

On the other hand, the exact values for the optimal constants $B_m^{\mathbb{K}}$ satisfying (1) are still unknown, although many progresses have been made in the last few years. Having nice estimates for these constants is, in general, crucial for applications (see [10]). The first estimates for $(B_m^{\mathbb{K}})_{m=1}^{\infty}$ [4,5,9,14] suggested an exponential growth but only few years ago very different results have appeared. In fact, for $m \geq 2$ the most recent estimates for the optimal values for the constants satisfying (1) show a sublinear growth:

$$B_m^{\mathbb{C}} < m^{0.21392}, \quad \text{and} \quad B_m^{\mathbb{R}} < 1.3 \cdot m^{0.36481}.$$

More specifically (see [3,6]),

$$1 \leq B_m^{\mathbb{C}} \leq \prod_{j=2}^m \Gamma\left(2 - \frac{1}{j}\right)^{\frac{j}{2-2j}}, \quad \text{and} \quad 2^{1-\frac{1}{m}} \leq B_m^{\mathbb{R}} \leq \prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1},$$

where Γ denotes the gamma function and the constants A_p denote the best constants satisfying Khinchine's inequality due to Haagerup (see [8]), which are given by

$$A_p := \sqrt{2} \left(\frac{\Gamma(\frac{p+1}{2})}{\sqrt{\pi}} \right)^{1/p}, \quad \text{for } p > p_0$$

$$A_p := 2^{\frac{1}{2} - \frac{1}{p}}, \quad \text{for } p \leq p_0.$$

Above, the number $p_0 \approx 1.85 \in (1, 2)$ is the solution of $\Gamma\left(\frac{p_0+1}{2}\right) = \frac{\sqrt{\pi}}{2}$. It is still an open problem, for real scalars, whether the optimal constants $B_m^{\mathbb{R}}$ are equal to $2^{1-\frac{1}{m}}$ or $\prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1}$ or whether they lie strictly between these bounds. The only known exact value appears in the real bilinear case, since $B_2^{\mathbb{R}} = 2^{1-\frac{1}{2}}$ (see [6]). For the complex case, similar (and harder) questions remain open.

In [1] it is proved that the Bohnenblust–Hille inequality, and the mixed (ℓ_1, ℓ_2) -Littlewood inequalities, are particular cases of a large family of sharp inequalities. More precisely, the following general result was proved in [1, Theorem 1.1] (the proof

uses the Kahane–Salem–Zygmund inequality (see [1]) and an interpolation argument which relies in the Hölder inequality for mixed sums):

Theorem 1.1 (Generalized Bohnenblust–Hille inequality, [1]) *Let $m \geq 2$ be a positive integer and let $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$. The following assertions are equivalent:*

(1) *The sequence $\mathbf{q}$ satisfies*

$$\frac{1}{q_1} + \dots + \frac{1}{q_m} \leq \frac{m+1}{2}.$$

(2) *There exists a constant $C_{m,\mathbf{q}}^{\mathbb{K}} \geq 1$ such that*

$$\left(\sum_{j_1=1}^{\infty} \left(\sum_{j_2=1}^{\infty} \left(\dots \left(\sum_{j_m=1}^{\infty} |T(e_{j_1}, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_2}{q_3}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \leq C_{m,\mathbf{q}}^{\mathbb{K}} \|T\|$$

for all continuous m -linear forms $T : c_0 \times \dots \times c_0 \rightarrow \mathbb{K}$.

See [15] for an even more general result.

Observe that the Bohnenblust–Hille inequality corresponds to the particular case $q_1 = q_2 = \dots = q_m = \frac{2m}{m+1}$, while the mixed (ℓ_1, ℓ_2) -Littlewood inequalities are the extreme cases in which $q_i = 1$ for some $i \in \{1, \dots, m\}$. In recent years, some works have provided optimal estimates for some particular constants $C_{m,\mathbf{q}}^{\mathbb{R}}$. For instance, $C_{2,\mathbf{q}}^{\mathbb{R}} = 2^{\frac{3}{2} - (\frac{1}{q_1} + \frac{1}{q_2})}$ for all $\mathbf{q} = (q_1, q_2)$, see [1], and $C_{m,\mathbf{q}}^{\mathbb{R}} = 2^{\frac{m-1}{2}}$, for $\mathbf{q} = (1, 2, 2, \dots, 2), \dots, (2, \dots, 2, 1)$, and all $m \geq 2$, see [12, 13].

This paper is organized as follows: In Sect. 2 we state and prove our main result related to the lower bounds of the constants in the Generalized Bohnenblust–Hille inequality and as consequence of our result we recover the optimal estimates recently obtained in [12, 13]. In Sect. 3 we improve some estimates for the upper bounds of that constants. The results of the Sects. 2 and 3 are used in the Sect. 4 to obtain sharp estimates in several situations. Additionally, in the final section, we state a conjecture about the optimal constants in the Generalized Bohnenblust–Hille inequality.

2 Main result

Let $T_2^{(1)} := T_2^{(2)} : c_0 \times c_0 \rightarrow \mathbb{R}$ given by

$$(x^{(1)}, x^{(2)}) \mapsto x_1^{(1)} x_1^{(2)} + x_1^{(1)} x_2^{(2)} + x_2^{(1)} x_1^{(2)} - x_2^{(1)} x_2^{(2)}.$$

Based on ideas from [13], we define recursively operators $T_m^{(k)}$ for all $m \geq 3$ and for all $1 \leq k \leq m$, starting with $k = m$:

$$\begin{aligned} T_m^{(m)} : c_0 \times \cdots \times c_0 &\rightarrow \mathbb{R} \\ (x^{(1)}, \dots, x^{(m)}) &\mapsto \left(x_1^{(m-1)} + x_2^{(m-1)} \right) T_{m-1}^{(m-1)} \left(x^{(1)}, \dots, x^{(m-2)}, x^{(m)} \right) \\ &\quad + \left(x_1^{(m-1)} - x_2^{(m-1)} \right) T_{m-1}^{(m-1)} \\ &\quad \times \left(x^{(1)}, \dots, x^{(m-2)}, B^{2^{m-2}} \left(x^{(m)} \right) \right) \end{aligned}$$

where B denotes the backward shift operator in c_0 . For $k < m$ we define

$$\begin{aligned} T_m^{(k)} : c_0 \times \cdots \times c_0 &\rightarrow \mathbb{R} \\ (x^{(1)}, \dots, x^{(m)}) &\mapsto \left(x_1^{(k+1)} + x_2^{(k+1)} \right) T_{m-1}^{(k)} \left(x^{(1)}, \dots, x^{(k)}, x^{(k+2)}, \dots, x^{(m)} \right) \\ &\quad + \left(x_1^{(k+1)} - x_2^{(k+1)} \right) T_{m-1}^{(k)} \left(x^{(1)}, \dots, B^{2^{m-2}} \left(x^{(k)} \right), x^{(k+2)}, \dots, x^{(m)} \right). \end{aligned}$$

For each m and each $1 \leq k \leq m$, it is readily seen that

$$\|T_m^{(k)}\| = 2^{m-1}.$$

Our main result is the following theorem that extends the main result of [12] and recovers [13, Theorem 3.1 and Corollary 3.2]:

Theorem 2.1 *Let $m \geq 2$ be a positive integer and $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$ be such that*

$$\frac{1}{q_1} + \cdots + \frac{1}{q_m} = \frac{m+1}{2}.$$

Then any constant $C_{m,\mathbf{q}}^{\mathbb{R}}$ such that

$$\left(\sum_{j_1=1}^{\infty} \left(\sum_{j_2=1}^{\infty} \left(\cdots \left(\sum_{j_m=1}^{\infty} |T(e_{j_1}, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_2}{q_3}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \leq C_{m,\mathbf{q}}^{\mathbb{R}} \|T\|$$

for all continuous m -linear forms $T : c_0 \times \cdots \times c_0 \rightarrow \mathbb{R}$ must satisfy

$$C_{m,\mathbf{q}}^{\mathbb{R}} \geq 2^{\frac{2m+3q_0-mq_0-4}{2q_0}}, \quad (5)$$

with $q_0 := \min \{q_1, \dots, q_m\}$.

Proof Since for each m and each $1 \leq k \leq m$, we know that

$$\|T_m^{(k)}\| = 2^{m-1},$$

and $\frac{1}{q_1} + \dots + \frac{1}{q_m} + \frac{m-2}{q_k} - (m-1) = \frac{2m+3q_k-mq_k-4}{2q_k}$, then it is sufficient to verify for each m and each $1 \leq k \leq m$ that

$$\left(\sum_{j_1=1}^{\infty} \left(\sum_{j_2=1}^{\infty} \left(\dots \left(\sum_{j_m=1}^{\infty} |T_m^{(k)}(e_{j_1}, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_2}{q_3}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} = 2^{\frac{1}{q_1} + \dots + \frac{1}{q_m} + \frac{m-2}{q_k}}.$$

The cases $m = 2$ and $k = 1, 2$ are immediate, because

$$\left(\sum_{i_1=1}^{\infty} \left(\sum_{i_2=1}^{\infty} \|T_2^{(k)}(e_{i_1}, e_{i_2})\|^{q_2} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} = 2^{\frac{1}{q_1} + \frac{1}{q_2}}.$$

Assume inductively that the result holds for $m-1$ and every $k \in \{1, \dots, m-1\}$. In order to simplify the notation, recall that the infinite sum of the respective $T_m^{(k)}$ is reduced to sums whose index j_i , for $i \neq k$, only change from 1 to 2. Thus, for $i \neq k$, we write $\sum_{j_i=1}^{\infty} = \sum_{j_i=1}^2 = \sum_{j_i}$. Furthermore $\sum_{j_k=1}^{\infty} = \sum_{j_k=1}^{2^{m-1}}$.

Let us begin with the case $k \neq m$.

$$\begin{aligned} & \sum_{j_k=1}^{\infty} \left(\sum_{j_{k+1}=1}^{\infty} \left(\sum_{j_{k+2}=1}^{\infty} \left(\dots \left(\sum_{j_m=1}^{\infty} |T_m^{(k)}(e_{j_1}, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\ &= \sum_{j_k=1}^{2^{m-1}} \left(\left(\sum_{j_{k+2}=1}^2 \left(\dots \left(\sum_{j_m=1}^2 |T_m^{(k)}(e_{j_1}, \dots, e_{j_k}, e_1, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\ & \quad + \left(\sum_{j_{k+2}=1}^2 \left(\dots \left(\sum_{j_m=1}^{2^{m-1}} |T_m^{(k)}(e_{j_1}, \dots, e_{j_k}, e_2, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \end{aligned}$$

$$\begin{aligned}
 &= \sum_{j_k=1}^{2^{m-1}} \left(\left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) + T_{m-1}^{(k)}(e_{j_1}, \dots, B^{2^{m-2}}(e_{j_k}), e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) - T_{m-1}^{(k)}(e_{j_1}, \dots, B^{2^{m-2}}(e_{j_k}), e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &= \sum_{j_k=1}^{2^{m-2}} \left(\left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \sum_{j_k=2^{m-2}+1}^{2^{m-1}} \left(\left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, B^{2^{m-2}}(e_{j_k}), e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, B^{2^{m-2}}(e_{j_k}), e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}}
 \end{aligned}$$

Observe that the last sum above can be rewritten as

$$\begin{aligned}
 &= \sum_{j_k=1}^{2^{m-2}} \left(\left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \sum_{j_k=1}^{2^{m-2}} \left(\left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &\quad + \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}} \\
 &= 2 \sum_{j_k=1}^{2^{m-2}} \left(2 \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_{k+1}}{q_{k+2}}} \right)^{\frac{q_k}{q_{k+1}}}
 \end{aligned}$$

$$= 2^{\frac{q_k}{q_{k+1}}} 2^{\sum_{j_k=1}^{2^{m-2}} \left(\sum_{j_{k+2}=1}^2 \left(\cdots \left(\sum_{j_m=1}^2 \left| T_{m-1}^{(k)}(e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_k}{q_{k+2}}}},$$

which implies

$$\begin{aligned} & \left(\sum_{j_1=1}^{\infty} \left(\cdots \left(\sum_{j_m=1}^{\infty} \left| T_m^{(k)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \\ &= \left(\sum_{j_1} \left(\cdots \sum_{j_{k-1}} \left(2^{\frac{q_k}{q_{k+1}}} 2^{\sum_{j_k=1}^{2^{m-2}} \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, \widehat{e_{j_k}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_k}{q_{k+2}}} \right)^{\frac{q_{k-1}}{q_k}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \\ &= 2^{\frac{1}{q_{k+1}}} 2^{\frac{1}{q_k}} \left(\sum_{j_1} \left(\cdots \sum_{j_{k-1}} \left(\sum_{j_k}^{2^{m-2}} \left(\sum_{j_{k+2}} \left(\cdots \left(\sum_{j_m} \left| T_{m-1}^{(k)}(e_{j_1}, \dots, \widehat{e_{j_k}}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \cdots \right)^{\frac{q_{k+2}}{q_{k+3}}} \right)^{\frac{q_k}{q_{k+2}}} \right)^{\frac{q_{k-1}}{q_k}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \\ &= 2^{\frac{1}{q_{k+1}}} 2^{\frac{1}{q_k} \left(\frac{1}{q_1} + \cdots + \frac{1}{q_k} + \frac{1}{q_{k+2}} + \cdots + \frac{1}{q_m} + \frac{m-3}{q_k} \right)} = 2^{\frac{1}{q_1} + \cdots + \frac{1}{q_m} + \frac{m-2}{q_k}} = 2^{\frac{m+1}{2} + \frac{m-2}{q_k}}. \end{aligned}$$

Here, we denote $\left(e_{j_1}, \dots, \widehat{e_{j_k}}, \dots, e_{j_m} \right) := (e_{j_1}, \dots, e_{j_k}, e_{j_{k+2}}, \dots, e_{j_m})$, and in the last line we use the induction hypothesis.

For $k = m$, we analogously prove that

$$\begin{aligned} & \left(\sum_{j_{m-2}=1}^{\infty} \left(\sum_{j_{m-1}=1}^{\infty} \left(\sum_{j_m=1}^{\infty} \left| T_m^{(m)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \right)^{\frac{q_{m-3}}{q_{m-2}}} \\ &= \left(\sum_{j_{m-2}} \left(\sum_{j_{m-1}} \left(\sum_{j_m=1}^{2^{m-1}} \left| T_m^{(m)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \right)^{\frac{q_{m-3}}{q_{m-2}}} \\ &= \left(\sum_{j_{m-2}} \left(\left(\sum_{j_m=1}^{2^{m-1}} \left| T_m^{(m)}(e_{j_1}, \dots, e_1, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} + \left(\sum_{j_m=1}^{2^{m-1}} \left| T_m^{(m)}(e_{j_1}, \dots, e_2, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \\ &= \left(\sum_{j_{m-2}} \left(\left(\sum_{j_m=1}^{2^{m-1}} \left| T_{m-1}^{(m-1)}(e_{j_1}, \dots, \widehat{e_{j_m}}, \dots, e_{j_m}) + T_{m-1}^{(m-1)}(e_{j_1}, \dots, e_{j_{m-1}}, B^{2^{m-2}}(e_{j_m})) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} + \left(\sum_{j_m=1}^{2^{m-1}} \left| T_{m-1}^{(m-1)}(e_{j_1}, \dots, \widehat{e_{j_m}}, \dots, e_{j_m}) - T_{m-1}^{(m-1)}(e_{j_1}, \dots, e_{j_{m-1}}, B^{2^{m-2}}(e_{j_m})) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \right)^{\frac{q_{m-3}}{q_{m-2}}} \end{aligned}$$

$$= \left(\sum_{j_{m-2}} \left(2 \left(2 \sum_{j_m=1}^{2^{m-2}} \left| T_{m-1}^{(m-1)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \right)^{\frac{q_{m-3}}{q_{m-2}}},$$

hence

$$\begin{aligned} & \sum_{j_1=1}^{\infty} \left(\dots \left(\sum_{j_{m-1}=1}^{\infty} \left(\sum_{j_m=1}^{\infty} \left| T_m^{(m)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \dots \right)^{\frac{q_k}{q_{k+1}}} \\ &= \sum_{j_1=1}^{\infty} \left(\dots \left(\sum_{j_{m-2}} \left(2 \left(2 \sum_{j_m=1}^{2^{m-2}} \left| T_{m-1}^{(m-1)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \right)^{\frac{q_{m-2}}{q_{m-1}}} \dots \right)^{\frac{q_{m-3}}{q_{m-2}}} \right)^{\frac{q_k}{q_{k+1}}} \\ &= 2^{\frac{1}{q_m}} 2^{\frac{1}{q_{m-1}}} \sum_{j_1} \left(\dots \left(\sum_{j_{m-2}} \left(\sum_{j_m=1}^{2^{m-2}} \left| T_{m-1}^{(m-1)}(e_{j_1}, \dots, e_{j_m}) \right|^{q_m} \right)^{\frac{q_{m-2}}{q_m}} \right)^{\frac{q_{m-3}}{q_{m-2}}} \dots \right)^{\frac{q_1}{q_2}} \\ &= 2^{\frac{1}{q_m}} 2^{\frac{1}{q_{m-1}}} 2^{\frac{1}{q_1} + \dots + \frac{1}{q_{m-2}} + \frac{1}{q_m} + \frac{m-3}{q_m}} = 2^{\frac{m+1}{2} + \frac{m-2}{q_m}} \end{aligned}$$

where in the last equality we use again the induction hypothesis, concluding thus the proof. $\square$

As an application of the above theorem we recover the optimal estimates recently obtained in [12, 13], in the mixed (ℓ_1, ℓ_2) -Littlewood inequalities:

Corollary 2.2 *Let $m \geq 2$ be a positive integer. For $1 \leq l \leq m$, let be $\mathbf{q} := (q_1, \dots, q_m)$ given by*

$$q_i = \begin{cases} 2, & \text{if } i \neq l; \\ 1, & \text{if } i = l. \end{cases}$$

Then $C_{m, \mathbf{q}}^{\mathbb{R}} = 2^{\frac{m-1}{2}}$.

Proof We know that

$$C_{m, \mathbf{q}}^{\mathbb{R}} \leq 2^{\frac{m-1}{2}}.$$

From the theorem above, $q_0 = 1$, we obtain $C_{m, \mathbf{q}}^{\mathbb{R}} \geq 2^{\frac{2m+3q_0-mq_0-4}{2q_0}} = 2^{\frac{m-1}{2}}$. $\square$

We observe that, in the above corollary, if exactly one entry q_i equals 1, then $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ has exponential growth. We can extend this result in the following sense:

Corollary 2.3 *Let $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$, such that $\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2}$. Let $l \in \mathbb{N}$ a fixed integer. If $\mathbf{q}$ is such that, for all $m > l$, the number of indices i such that $q_i = \frac{2l}{l+1}$ is l , then $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ has exponential growth.*

Proof Let $m > l$, $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$, such that $\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2}$, and $I_{l,m} \subset \{1, \dots, m\}$ the set of indices i such that $q_i = \frac{2l}{l+1}$. Obviously, if $i \in \{1, \dots, m\} \setminus I_{l,m}$, then $q_i = 2$.

We apply (5) with $q_0 = \frac{2l}{l+1}$, and then we have

$$\begin{aligned} C_{m,\mathbf{q}}^{\mathbb{R}} &\geq 2^{\frac{2m+3q_0-mq_0-4}{2q_0}} \\ &= 2^{\frac{2m+3\left(\frac{2l}{l+1}\right)-m\left(\frac{2l}{l+1}\right)-4}{2\left(\frac{2l}{l+1}\right)}} \\ &= 2^{\frac{l+m-2}{2l}}. \end{aligned}$$

Since l is fixed, we conclude that $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ has exponential growth. $\square$

Remark 2.4 According with [13, Sect. 3], let $\text{diam}(\mathbf{q}) = \text{diam}(q_1, \dots, q_m) := \max |q_i - q_j|$. Recall that if $\mathbf{q}$ is the multiexponent in the classical Bohnenblust–Hille inequality, then $\text{diam}(\mathbf{q}) = 0$ and $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ is sublinear. On the other hand, if $\mathbf{q}$ is a multiexponent in any mixed (ℓ_1, ℓ_2) -Littlewood inequality, then $\text{diam}(\mathbf{q}) = 1$ and $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ is exponential. Finally, if $\mathbf{q}$ satisfies the conditions of the above corollary, $\text{diam}(\mathbf{q}) = \frac{2}{l+1}$, and $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ is exponential. We observe that $\lim_{l \rightarrow \infty} \frac{2}{l+1} = 0$. Thus, we obtain a family of multiexponents $\mathbf{q}$ with $\text{diam}(\mathbf{q})$ arbitrarily small, such that $(C_{m,\mathbf{q}}^{\mathbb{R}})_m$ is exponential.

Remark 2.5 If $\mathbf{q} = \left(\frac{2m}{m+1}, \dots, \frac{2m}{m+1}\right)$ is the exponent in the classical Bohnenblust–Hille inequality, then by applying Theorem 2.1 we recover the lower estimates given in [6]. In fact

$$\begin{aligned} B_m^{\mathbb{R}} &= C_{m,\mathbf{q}}^{\mathbb{R}} \\ &\geq 2^{\frac{2m+3q_0-mq_0-4}{2q_0}} \\ &= 2^{\frac{2m+3\left(\frac{2m}{m+1}\right)-m\left(\frac{2m}{m+1}\right)-4}{2\left(\frac{2m}{m+1}\right)}} \\ &= 2^{1-\frac{1}{m}}. \end{aligned}$$

3 Some remarks on the upper estimates of the general Bohnenblust–Hille inequality

In this section we extend some recent results providing upper estimates for the constants in the Generalized Bohnenblust–Hille inequality. These results will be used in the following section. We begin by recalling two results:

Lemma 3.1 ([2, Lemma 2.1]) *Let $(q_1, \dots, q_m) \in [1, 2]^m$ such that $\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2}$. If $q_i \geq \frac{2m-2}{m}$ for some index i and $q_k = q_l$ for all $k \neq i$ and $l \neq i$, then*

$$C_{m, \mathbf{q}}^{\mathbb{K}} \leq \prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1},$$

where $A_{\frac{2j-2}{j}}$ are the constants of Khinchine's inequality.

Theorem 3.2 ([2, Theorem 2.3]) *If $m \geq 2$ is a positive integer, and $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$ are such that*

$$\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2},$$

and

$$\max_i q_i < \frac{2m^2 - 4m + 2}{m^2 - m - 1}$$

then

$$C_{m, \mathbf{q}}^{\mathbb{K}} \leq \prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1}.$$

Using Lemma 3.1 and Theorem 3.2 together with an interpolative approach (see [1, Proposition 2.1]) we can prove the following slightly more general result:

Proposition 3.3 *Let $m \geq 2$ and N be positive integers and $\mathbf{q}, \mathbf{q}(1), \dots, \mathbf{q}(N) \in [1, 2]^m$ be such that $\left(\frac{1}{q_1}, \dots, \frac{1}{q_m}\right)$ belongs to the convex hull of the tuples $\left(\frac{1}{q_1(k)}, \dots, \frac{1}{q_m(k)}\right)$, with $k = 1, \dots, N$. Suppose that for each k we have*

$$\frac{1}{q_1(k)} + \dots + \frac{1}{q_m(k)} = \frac{m+1}{2},$$

and that either $\max_i q_i(k) < \frac{2m^2-4m+2}{m^2-m-1}$, or $q_i(k) \geq \frac{2m-2}{m}$ for some i and $q_j(k) = q_l(k)$ for all $j \neq i$ and $l \neq i$. Then we have

$$C_{m, \mathbf{q}}^{\mathbb{K}} \leq \prod_{t=2}^m A_{\frac{2t-2}{t}}^{-1}.$$

Proof If $\max_i q_i(k) < \frac{2m^2-4m+2}{m^2-m-1}$ then from Lemma 3.1 we have

$$C_{m,\mathbf{q}(k)}^{\mathbb{K}} \leq \prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1}.$$

In the other case, if $q_i(k) \geq \frac{2m-2}{m}$ for some index i and $q_j(k) = q_l(k)$ for all $j \neq i$ and $l \neq i$, from Theorem 3.2 we obtain

$$C_{m,\mathbf{q}(k)}^{\mathbb{K}} \leq \prod_{j=2}^m A_{\frac{2j-2}{j}}^{-1}.$$

By hypothesis $\mathbf{q} = \left(\frac{1}{q_1}, \dots, \frac{1}{q_m}\right)$ belongs to the convex hull of the points $\left(\frac{1}{q_1(k)}, \dots, \frac{1}{q_m(k)}\right)$ with $k = 1, \dots, N$, i. e.

$$\frac{1}{q_i} = \frac{\theta_1}{q_i(1)} + \dots + \frac{\theta_N}{q_i(N)},$$

with $\sum_{k=1}^N \theta_k = 1$ and $\theta_k \in [0, 1]$ for all k . The interpolation procedure from [1, Proposition 2.1] yields

$$\|T(e_{j_1}, \dots, e_{j_m})\|_{\mathbf{q}} \leq \prod_{k=1}^N \|T(e_{j_1}, \dots, e_{j_m})\|_{\mathbf{q}(k)}^{\theta_k}.$$

From Lemma 3.1 and/or Theorem 3.2 we have

$$\begin{aligned} \prod_{k=1}^N \|T(e_{j_1}, \dots, e_{j_m})\|_{\mathbf{q}(k)}^{\theta_k} &\leq \prod_{k=1}^N \left(\prod_{t=2}^m A_{\frac{2t-2}{t}}^{-1} \|T\| \right)^{\theta_k} \\ &= \prod_{t=2}^m A_{\frac{2t-2}{t}}^{-1} \|T\|, \end{aligned}$$

hence

$$\begin{aligned} \|T(e_{j_1}, \dots, e_{j_m})\|_{\mathbf{q}} &= \left(\sum_{j_1=1}^n \left(\sum_{j_2=1}^n \left(\dots \left(\sum_{j_m=1}^n |T(e_{j_1}, \dots, e_{j_m})|^{q_m} \right)^{\frac{q_{m-1}}{q_m}} \dots \right)^{\frac{q_2}{q_3}} \right)^{\frac{q_1}{q_2}} \right)^{\frac{1}{q_1}} \\ &\leq \prod_{t=2}^m A_{\frac{2t-2}{t}}^{-1} \|T\|. \end{aligned}$$

□

Example 3.4 Note that the above result encompasses cases not covered by the previous results, and we still have

$$C_{m,\mathbf{q}}^{\mathbb{K}} \leq \prod_{t=2}^m A_{\frac{2t-2}{t}}^{-1}.$$

For instance, let $m = N = 3$ and $\mathbf{q}(1) = (2, \frac{4}{3}, \frac{4}{3})$, $\mathbf{q}(2) = (\frac{4}{3}, 2, \frac{4}{3})$, $\mathbf{q}(3) = (\frac{4}{3}, \frac{4}{3}, 2)$. Applying Proposition 3.3 to

$$\mathbf{q} = \left(\frac{4}{3-\theta_1}, \frac{4}{3-\theta_2}, \frac{4}{3-\theta_3} \right); \theta_1, \theta_2, \theta_3 \in [0, 1] \text{ and } \theta_1 + \theta_2 + \theta_3 = 1,$$

we obtain $C_{3,\mathbf{q}}^{\mathbb{K}} \leq \prod_{t=2}^3 A_{\frac{2t-2}{t}}^{-1}$.

Choosing $(\theta_1, \theta_2, \theta_3) = (\frac{7}{10}, \frac{1}{10}, \frac{2}{10})$ we get

$$\mathbf{q} = \left(\frac{40}{23}, \frac{40}{29}, \frac{40}{28} \right),$$

which does not satisfy neither hypotheses of Lemma 3.1, nor the hypotheses of Theorem 3.2 (because $\frac{40}{23} > \frac{2(3)^2-4(3)+2}{(3)^2-(3)-1} = \frac{8}{5}$).

4 Application: sharp estimates for the generalized Bohnenblust–Hille inequality

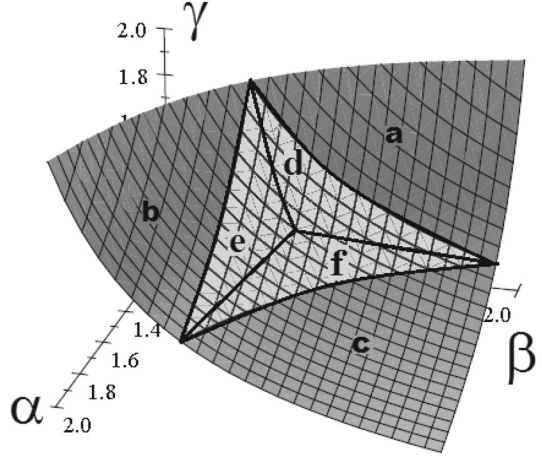
In this section we use the results of the previous sections to obtain sharp estimates for several multiexponents of the Generalized Bohnenblust–Hille inequality for m -linear forms. We start with the case $m = 3$. Recall that the sharp estimates for the multiexponents $\mathbf{q} = (1, 2, 2)$, $(2, 1, 2)$, $(2, 2, 1)$ have already been shown in [12, 13].

Theorem 4.1 Let $\mathbf{q} := (\alpha, \beta, \gamma) \in [1, 2]^3$ be such that $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$, and $q_0 := \min \{\alpha, \beta, \gamma\}$. If $q_0 \leq \frac{4}{3}$, then the optimal constant $C_{3,\mathbf{q}}^{\mathbb{R}}$ is $2^{\frac{1}{q_0}}$.

Proof If $\mathbf{q} := (\alpha, \beta, \gamma)$ is according with the hypothesis of the theorem, then there are $\tau \in [0, 1]$ and $\theta \in [0, 1]$, such that

$$\begin{aligned} (a) \quad \mathbf{q} &= \left(\frac{4}{\theta+3}, \frac{4}{2+\tau-\theta\tau}, \frac{4}{3+\theta\tau-\theta-\tau} \right), \quad \text{or} \\ (b) \quad \mathbf{q} &= \left(\frac{4}{3+\theta\tau-\theta-\tau}, \frac{4}{\theta+3}, \frac{4}{2+\tau-\theta\tau} \right), \quad \text{or} \\ (c) \quad \mathbf{q} &= \left(\frac{4}{2+\tau-\theta\tau}, \frac{4}{3+\theta\tau-\theta-\tau}, \frac{4}{\theta+3} \right) \end{aligned}$$

Fig. 1 The 3-linear case: the graph, centered at $(1, 1, 1)$, represents the points such that $(\alpha, \beta, \gamma) \in [1, 2]^3$ with $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$. The darker region corresponds to points $\mathbf{q}$ with optimal constants (Theorem 4.1); the pale region corresponds to the situation of Theorem 5.2



(in order to a better comprehension see Fig. 1). Since $\frac{4}{\theta+3} \leq \frac{4}{2+\tau-\theta\tau}$ and $\frac{4}{\theta+3} \leq \frac{4}{3+\theta\tau-\theta-\tau}$, it follows from Theorem 2.1 (with $m = 3$) that

$$C_{3,\mathbf{q}}^{\mathbb{R}} \geq 2^{\frac{\theta+3}{4}}.$$

On the other hand, we need to show that if $\mathbf{q}$ is of the form (a), (b) or (c), then

$$C_{3,\mathbf{q}}^{\mathbb{R}} \leq 2^{\frac{\theta+3}{4}}.$$

Suppose $\mathbf{q}$ as in (a). If $\theta = 0$, from [2, Lemma 2.1] we know that for $\mathbf{q}(1) = (\frac{4}{3}, 2, \frac{4}{3})$, $\mathbf{q}(2) = (\frac{4}{3}, \frac{4}{3}, 2)$, we have $C_{3,\mathbf{q}(1)}^{\mathbb{R}} \leq 2^{\frac{3}{4}}$ and $C_{3,\mathbf{q}(2)}^{\mathbb{R}} \leq 2^{\frac{3}{4}}$, and since $\mathbf{q} = (\frac{4}{3}, \frac{4}{2+\tau}, \frac{4}{3-\tau})$ belongs to the convex hull of $\mathbf{q}(1)$ and $\mathbf{q}(2)$ for $\tau \in [0, 1]$, from Proposition 3.3 with $k = 2$ we conclude that

$$C_{3,\mathbf{q}}^{\mathbb{R}} \leq 2^{\frac{3}{4}}.$$

Let us prove the case $\theta \in (0, 1]$. From [12, Theorem 2.1] we know that for $\bar{\mathbf{q}}(1) = (1, 2, 2)$ we have $C_{3,\bar{\mathbf{q}}(1)}^{\mathbb{R}} = 2$. Moreover, from what we just did we have, for $\tau \in [0, 1]$ and $\bar{\mathbf{q}}(2) = (\frac{4}{3}, \frac{4}{2+\tau}, \frac{4}{3-\tau})$, that $C_{3,\bar{\mathbf{q}}(2)}^{\mathbb{R}} = 2^{\frac{3}{4}}$. Interpolating $\bar{\mathbf{q}}(1) = (1, 2, 2)$ and $\bar{\mathbf{q}}(2) = (\frac{4}{3}, \frac{4}{2+\tau}, \frac{4}{3-\tau})$ we obtain

$$\begin{aligned} \frac{1}{q_1} &= \frac{\theta}{1} + \frac{1-\theta}{\frac{4}{3}} \Rightarrow q_1 = \frac{4}{\theta+3}; \\ \frac{1}{q_2} &= \frac{\theta}{2} + \frac{1-\theta}{\frac{4}{2+\tau}} \Rightarrow q_2 = \frac{4}{2+\tau-\theta\tau}; \end{aligned}$$

$$\frac{1}{q_3} = \frac{\theta}{2} + \frac{1-\theta}{\frac{4}{3-\tau}} \Rightarrow q_3 = \frac{4}{3 + \theta\tau - \tau - \theta}.$$

Thus, $\mathbf{q}$ belongs to the convex hull of $\bar{\mathbf{q}}(1)$ and $\bar{\mathbf{q}}(2)$, which implies

$$C_{3,\bar{\mathbf{q}}}^{\mathbb{R}} \leq 2^\theta 2^{\frac{3}{4}(1-\theta)} = 2^{\frac{\theta+3}{4}}.$$

We conclude then that

$$C_{3,\mathbf{q}}^{\mathbb{R}} = 2^{\frac{\theta+3}{4}}$$

for $\mathbf{q} = \left(\frac{4}{\theta+3}, \frac{4}{2+\tau-\theta\tau}, \frac{4}{3+\theta\tau-\theta-\tau} \right)$ with $\tau \in [0, 1]$ and $\theta \in [0, 1]$. This concludes the proof in the case that (a) occurs. The proof in the cases that $\mathbf{q}$ is like (b), or (c) is analogous. $\square$

Using the ideas from the proof above we can prove the following more general result:

Theorem 4.2 *Let $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$ such that $\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2}$, and $q_0 := \min \{q_1, \dots, q_m\}$. If $q_0 \leq \frac{4}{3}$, then the optimal constant $C_{m,\mathbf{q}}^{\mathbb{R}}$ is $2^{\frac{2m+3q_0-mq_0-4}{2q_0}}$.*

5 On the conjecture of Pellegrino–Teixeira

In the paper [11, Problem 11.1] it was posed the following open problem that questions the existence of a strong Bohnenblust–Hille inequality:

Problem 5.1 Is there a universal constant $K_{\mathbb{K}}$ so that

$$\left(\sum_{i_1, \dots, i_m=1}^{\infty} \left| U(e_{i_1}, \dots, e_{i_m}) \right|^{\frac{2m}{m+1}} \right)^{\frac{m+1}{2m}} \leq K_{\mathbb{K}} \|U\|$$

for every positive integer $m \geq 1$ and all continuous m -linear forms $U : c_0 \times \dots \times c_0 \rightarrow \mathbb{K}$?

The possibility of having a positive answer, at least for the real case, has recently taken hold. In fact, motivated by the results obtained in the paper [13], the authors further made the next conjecture:

Conjecture 5.1 *The optimal constants in the Bohnenblust–Hille inequality are universally bounded, irrespectively of the value of m . In the real case, the best constants should be precisely $2^{1-\frac{1}{m}}$.*

Let us recall, we saw previously that if we apply Theorem 2.1 with the exponent $\mathbf{q} = \left(\frac{2m}{m+1}, \dots, \frac{2m}{m+1} \right)$, the estimate $2^{1-\frac{1}{m}}$ is exactly obtained. In addition, when

Theorem 2.1 was applied in the case of the mixed (ℓ_1, ℓ_2) -Littlewood inequality, the optimal estimate $2^{\frac{m-1}{2}}$ was obtained (Corollary 2.2). Then, if Conjecture 5.1 is true, Theorem 2.1 would bring optimal estimates for the extremal cases in the Generalized Bohnenbust–Hille inequality, i.e., the mixed (ℓ_1, ℓ_2) -Littlewood inequality and the classic Bohnenbust–Hille inequality. Furthermore, if Conjecture 5.1 is true for $m = 3$, it means if $B_3^{\mathbb{R}} = 2^{\frac{2}{3}}$, then by using an argument similar to the proof of Theorem 4.1, the missing optimal estimates in Fig. 1 can be obtained:

Theorem 5.2 Let $\mathbf{q} := (\alpha, \beta, \gamma) \in [1, 2]^3$ such that $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 2$, and $q_0 := \min\{\alpha, \beta, \gamma\}$. If $B_3^{\mathbb{R}} = 2^{\frac{2}{3}}$ then the optimal constant $C_{3,\mathbf{q}}^{\mathbb{R}}$ is $2^{\frac{1}{q_0}}$.

Proof If $q_0 \leq \frac{4}{3}$, the result is true without the assumption $B_3^{\mathbb{R}} = 2^{\frac{2}{3}}$ (see Theorem 4.1). Otherwise there exist $\tau, \theta \in [0, 1]$ such that

$$\begin{aligned} (d) \quad \mathbf{q} &= \left(\frac{12}{9-\theta}, \frac{12}{2\theta+3\tau-3\theta\tau+6}, \frac{12}{3\theta\tau+9-\theta-3\tau} \right), \quad \text{or} \\ (e) \quad \mathbf{q} &= \left(\frac{12}{3\theta\tau+9-\theta-3\tau}, \frac{12}{9-\theta}, \frac{12}{2\theta+3\tau-3\theta\tau+6} \right), \quad \text{or} \\ (f) \quad \mathbf{q} &= \left(\frac{12}{2\theta+3\tau-3\theta\tau+6}, \frac{12}{3\theta\tau+9-\theta-3\tau}, \frac{12}{9-\theta} \right) \end{aligned}$$

(see Fig. 1). Since $\frac{12}{9-\theta} \leq \frac{12}{2\theta+3\tau-3\theta\tau+6}$ and $\frac{12}{9-\theta} \leq \frac{12}{3\theta\tau+9-\theta-3\tau}$, by Theorem 2.1 we have

$$C_{3,\mathbf{q}}^{\mathbb{R}} \geq 2^{\frac{9-\theta}{12}}.$$

Thus we need to show that if $\mathbf{q}$ is of the form (d), (e) or (f), then

$$C_{3,\mathbf{q}}^{\mathbb{R}} \leq 2^{\frac{9-\theta}{12}}.$$

Suppose $\mathbf{q}$ as in (d). Let $\mathbf{q}(0) = (\frac{3}{2}, \frac{3}{2}, \frac{3}{2})$, and $\mathbf{q}(1) = (\frac{4}{3}, \frac{4}{2+\tau}, \frac{4}{3-\tau})$. Interpolating

$$\begin{aligned} \frac{1}{q_1} &= \frac{2\theta}{3} + \frac{1-\theta}{\frac{4}{3}} \Rightarrow q_1 = \frac{12}{9-\theta} \\ \frac{1}{q_2} &= \frac{2\theta}{3} + \frac{1-\theta}{\frac{4}{2+\tau}} \Rightarrow q_2 = \frac{12}{2\theta+3\tau-3\theta\tau+6} \\ \frac{1}{q_3} &= \frac{2\theta}{3} + \frac{1-\theta}{\frac{4}{3-\tau}} \Rightarrow q_3 = \frac{12}{3\theta\tau+9-\theta-3\tau} \end{aligned}$$

thus, $\mathbf{q}$ belongs to the convex hull of $\mathbf{q}(0)$ and $\mathbf{q}(1)$, which implies

$$\begin{aligned} C_{3,\mathbf{q}}^{\mathbb{R}} &\leq \left(2^{\frac{2}{3}}\right)^{\theta} \left(2^{\frac{3}{4}}\right)^{1-\theta} \\ &= 2^{\frac{9-\theta}{12}} \end{aligned}$$

and this concludes the proof in the case (d). The proof in the cases (e) or (f) is analogous. $\square$

We believe that Conjecture 5.1 is true. Moreover, based in Theorem 5.2, we believe that the following more general conjecture, that connects Conjecture 5.1 with the optimal estimates of the last section, is true:

Conjecture 5.2 *Let $\mathbf{q} := (q_1, \dots, q_m) \in [1, 2]^m$ be such that $\frac{1}{q_1} + \dots + \frac{1}{q_m} = \frac{m+1}{2}$, and $q_0 := \min \{q_1, \dots, q_m\}$. Then the optimal constant $C_{m,\mathbf{q}}^{\mathbb{R}}$ is $2^{\frac{2m+3q_0-mq_0-4}{2q_0}}$.*

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